

Ques:- The State of Stress at a point is given by the matrix

$$\tau = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

Determine the principal Stress and the principal planes on which the principal Stresses will act.

Soln:- The characteristic Eqn is given by
 $(\tau - \sigma I)\hat{n} = 0$

$$\begin{vmatrix} 6-\sigma & -2 & 2 \\ -2 & 3-\sigma & -1 \\ 2 & -1 & 3-\sigma \end{vmatrix} = 0$$

Upon few algebraic manipulations

$$(\sigma - 2)(\sigma - 2)(\sigma - 8) = 0$$

$\therefore$ The principal Stresses are $\sigma_1 = 8, \sigma_2 = \sigma_3 = 2$

The characteristic vector of τ corresponding to the characteristic root '8' is given by the non-zero soln of the equation.

$$\begin{bmatrix} 6-8 & -2 & 2 \\ -2 & 3-8 & -1 \\ 2 & -1 & 3-8 \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

After few row manipulations we get-

$$\begin{bmatrix} -2 & -2 & 2 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Note:- The Coefficient matrix of these Equations is of
 RANK-2 (Rank = No. of non zero rows in Coeff matrix)
 $\therefore$ These equations possess
 $n-R = 3-2 = 1$ linearly independent solution.

The aforesaid matrix can be written in the Equation form as

$$\left. \begin{aligned} -2n_1 - 2n_2 + 2n_3 &= 0 \\ -3n_2 - 3n_3 &= 0 \end{aligned} \right\}$$

From Last Eqn

$$n_2 = -n_3$$

$$\text{let } n_3 = 1 \Rightarrow n_2 = -1$$

$$\& \text{ From 1st Eqn. } n_1 = n_3 - n_2 = 1 - (-1) = 2$$

$\therefore$ The vector

$$n^{(1)} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

is an eigen vector of the stress matrix.
 Corresponding to $\sigma = 8 \text{ MPa}$.
 $\rightarrow$ Finally it has to be normalised to unit vector
 (Do it yourself)

Now:- The eigen vector of τ corresponding to $\sigma = 2$ are
 given by the non-zero solution of the Eqn.

$$(\tau - 2I) \hat{n} = 0$$

$$\begin{bmatrix} 4 & -2 & -2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

After few row manipulations it yields

$$\begin{bmatrix} -2 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Rank of the Coeff. matrix = 1 (= No. of non-zero rows)

$\therefore$ These Eqns possess $n-R = 3-1 = 2$ linearly independent
 Soln:

The system reduces to

$$-2n_1 + n_2 - n_3 = 0$$

$$\text{or, } n_2 = n_3 + 2n_1$$

one vector is $n_1 = 0, n_3 = 1 \Rightarrow n_2 = 1$

$$n^{(2)} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

we see that $n^{(1)} \cdot n^{(2)} = 0$
 $\Rightarrow n^{(1)} \perp n^{(2)}$

Now we seek to find the third vector $n^{(3)}$ orthogonal to $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$
under the constraint $n_2 = n_3 + 2n_1$.

The generalised vector with such constraint could be

$$(n_1, n_3 + 2n_1, n_3)$$

So as per constraint

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} n_1 \\ n_3 + 2n_1 \\ n_3 \end{bmatrix} = 0 \quad [\text{orthogonality}]$$

$$\Rightarrow 0n_1 + (n_3 + 2n_1) \times 1 + n_3 \times 1 = 0$$

$$\Rightarrow n_3 + 2n_1 + n_3 = 0$$

$$\text{or, } n_1 + n_3 = 0$$

$$\therefore \forall n_1 = 1 \Rightarrow n_3 = -1$$

$$\& n_2 = n_3 + 2n_1 = -1 + 2 \times 1 = 1$$

$$\therefore \text{The third vector is } \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = n^{(3)}$$

Note:-

Corresponding to the eigen values $\sigma_2 = 2$ & $\sigma_3 = 2$

The orthogonal vectors are $n^{(2)} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ and $n^{(3)} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$

if c_1 & c_2 are scalars not both equal to zero then

$c_1 n^{(2)} + c_2 n^{(3)}$ gives all eigenvectors corresponding to $\sigma = 2$.

$\Rightarrow \sigma_1$ will be fixed and σ_2 can have any direction in the plane $\perp$ to σ_1

